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Lora-based IoT sensor node for Real-time Flood Early Warning System

Yoeseph, N.M.^{1,*}, Purnomo, F.A.¹, Hartono, R.¹, and Nuryani²

¹ Informatics Engineering Departement, Sebelas Maret University

² Physisc Departement, Sebelas Maret University

*Email: nanang.my@staff.uns.ac.id

Abstract. Internet of Things (IoT) using LoRa technology has been widely used in many fields. One of them is in disaster mitigation and management. This paper presents sensor nodes as part of a Flood Early Warning System using LoRa technology. The node consists of a water level sensor and a water flow meter. Water level measurements were carried out by ultrasonic sensors HY-SRF05, and the speed of water flow is measured by water flow meter YF-S201. Communication between the nodes and the gateway is done by the BsFrance Lora32u4 II v1.4 Wireless Transceiver module. Data regarding the height of the water level and speed of water flow will be measured by the sensor and sent by the LoRa device to the gateway. The gateway then forwards that data to an open-source IoT platform for analysis and visualization. Testing ultrasonic sensor readings get an accuracy of 97.06%. Waterflow sensor testing obtained 91.96% accuracy. In addition to testing the measurement accuracy, data transmission consistency is also tested against the distance between the gateway and the node. The test results show that the data can still be sent at a space between the node and the gateway below 400 m. Above 400m, the data cannot be received by the gateway anymore. Testing delay time between reading data and receiving data on the IoT platform shows the average delivery delay of 0 ms.

1. Introduction

The flood phenomenon occurs in areas around rivers or urban areas, creating an early warning system. A Flood Early Warning System was designed, consisting of sensor nodes, LoRa gateway, cloud server, and user interface application. Sensor nodes are used for measuring the water level and the water flow and can send data for analysis and alerts. LoRa is a proprietary spread spectrum modulation scheme derivative of Chirp Spread Spectrum modulation (CSS) [1].

LoRa is a technology module that is different from others, such as wifi, GSM, Zigbee, and Bluetooth. In addition, LoRa has less power than Mosul GSM/LTE and has a more extended range of up to 10 Km than wifi, Bluetooth, and Zigbee [2]. In Lora-based Internet of Things system, when the sensor on the node detects data, it passes through the gateway and server for data transmission [3]. The End-note, which has many sensors, is placed within the LoRa range to monitor an area. The acquired data is then transmitted wirelessly via a LoRa gateway device connected to a cloud server [2].

Due to those characteristics, Lora technology has been used in several fields, such as agriculture[4][5][6] and weather stations[2][7]. Several Flood Early Warning Systems using LoRa technology had been published [8][9]. This paper will focus on the construction of Lora Sensor Nodes as part of Flood EWS. The Sensor Nodes consist of a water level meter and water flow meter. The



experiment is done by testing the accuracy of each sensor. In addition, a latency test is also carried out to test the consistency of data transmission against the distance between the gateway and the node.

2. Methods

The early warning system architecture consists of a LoRa node connected to several sensors, a LoRa gateway, a LoRAWAN network server, and a database. A LoRa gateway is used to bridge the LoRa wireless network to an IP network through wifi and ethernet [10]. The system uses The Things Network (TTN) as a LoRAWAN network server. The Things Network is an open-source infrastructure that provides open tools and networks for building IoT applications. Data from the sensors is then saved in the database through HTTP integration with The Things Network. The EWS system architecture is shown in Figure 1.

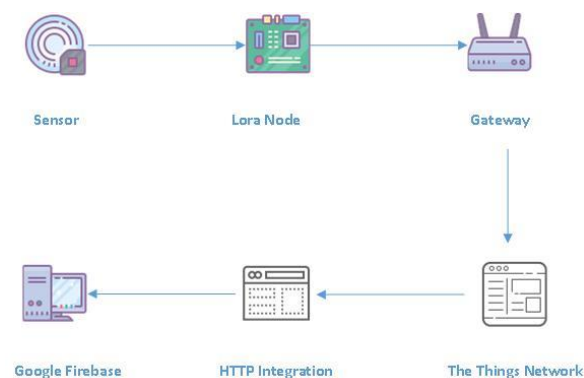


Figure 1. EWS System Architecture

LoRa nodes are built with BSFrance Lora32u4 v1.4 as LoRA wireless module to transmit data from ultrasonic sensors and water flow meters. The ultrasonic sensor is used to monitor water level, and the water flow meter is used to monitor the speed of water flow. The ultrasonic sensor used is HY-SRF05 module. The water flow sensor used is YF-S201.

Ultrasonic sensors work by utilizing waves ultrasonics. Ultrasonic is a wave whose frequency is above the sound wave. Ultrasonic sensors used in this research are the SRF05 module which is shown in Figure 2. This sensor works using a transmitter that emits ultrasonic waves and a receiver that receives back waves ultrasonics. This HY-SRF05 ultrasonic sensor works on a DC voltage of 5 Volts, a current load of 30 mA to 50 mA, with a frequency wave of 40 kHz. This sensor can read distances between 3 cm to 400 cm.



Figure 2. HY-SRF05 Ultrasonic Sensor

Waterflow is a sensor used to measure the flow of water flowing flow. This sensor has a pinwheel inside, connected with an integrated magnetic hall effect sensor that emits an electrical pulse for each round. By counting the vibration from the sensor output, the water flow can be estimated. The water flow sensor model used for this research is YF-S201 which is shown in Figure 3. This sensor model works on a voltage of 5 to 18V DC, with 5V output. This sensor can work with flow speed up to 30 liters/minute, at a temperature of -25 to 80°C and a humidity of 35% - 80% RH. The accuracy of this sensor ranges from 10%. The maximum acceptable water pressure is 2 MPa.



Figure 3. YF-S201 Water Flow Sensor

Ultrasonic sensor accuracy test is done by measuring the distance captured by the sensor with the existing accurate length. As for testing of the water flow sensor is done by testing the current water flow with the amount of water obtained in 60 seconds. Latency is measured by calculating the time between the data read by the sensor and the data saved in the database.

3. Result and Discussion

Testing on the node focuses on sensor accuracy and the time difference between sensor readings and data received by the database. The accuracy of the ultrasonic sensor is determined by comparing the distance data obtained by the sensor with the actual distance. The ultrasonic sensor SRF05 accuracy test result is shown in Figure 4. From the figure, it can be seen that the accuracy obtained from ultrasonic sensor testing is 97.06%.

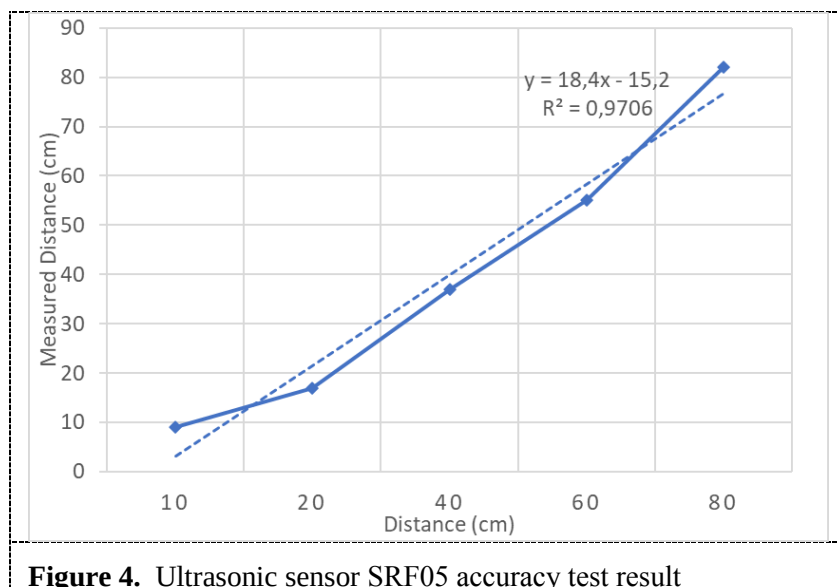


Figure 4. Ultrasonic sensor SRF05 accuracy test result

The testing of the water flow sensor is carried out by connecting it to a faucet that can be adjusted to the amount of water that flows out. The amount of water in 60 seconds is then calculated. The accuracy of the water flow sensor is obtained by comparing the value from the sensor with the volume of water obtained. The water flow sensor YF-S201 accuracy-test can be shown in Figure 5. From the figure, it can be seen that the accuracy obtained from water flow sensor testing is 91.96%.

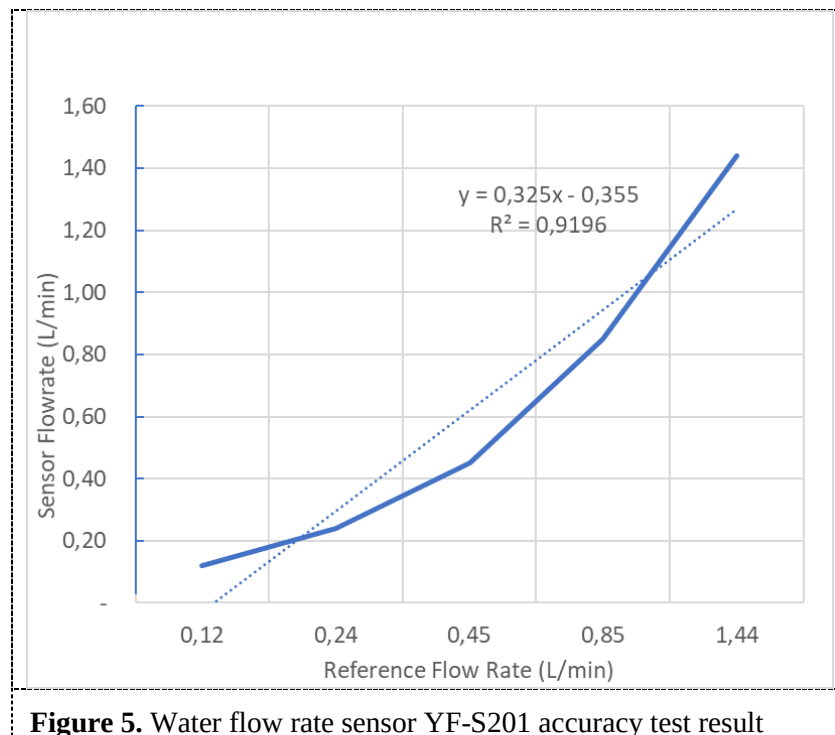


Figure 5. Water flow rate sensor YF-S201 accuracy test result

The latency test calculates the time between the data read by the sensor and sent by the nodes with the data received in the databases. This data latency is essential to know whether data arrive immediately at the database after the sensor has collected the data. Data transmission time is obtained from the time of sensor readings from the Serial monitor. The time of receiving the data is obtained from the timestamp of the data obtained in the Firebase console. Data latency is shown in Table 1. In Table 1, the latency between data transmits, and data receive 0 seconds. Zero-data latency can be called real-time data. The nodes for the Early Warning System can be used to monitor water levels and water flow rates. The transmitted data can be classified as zero-data latency or so-called real-time data.

Table 1. Data latency

Data Transmit (UTC+7)	Data Receive (UTC+7)	Latency (sec)
16:25:56	16:25:56	0
02:30:00	02:30:00	0
02:49:58	02:49:58	0
02:58:34	02:58:35	0
03:11:56	03:11:56	0

4. Conclusions

The conclusion of this paper presented is a manufacture of sensor nodes as part of a Flood Early Warning System using LoRa technology. The ultrasonic sensor readings test gets an accuracy of 97.06%, and water flow sensor testing obtained 91.96% accuracy. The delay time between reading data and receiving data on the IoT platform shows the delivery delay of 0 ms.

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